

3.4 The Derivative as a Rate of Change

In this section we study applications in which derivatives model the rates at which things change. It is natural to think of a quantity changing with respect to time, but other variables can be treated in the same way.

① Instantaneous Rates of Change

If we interpret the difference quotient

$$\frac{f(x+h) - f(x)}{h}$$

as the average rate of change in f over the interval from x to $x+h$, we can interpret its limit as $h \rightarrow 0$ as the instantaneous rate at which f is changing at the point x . This gives an important interpretation of the derivative.

DEFINITION The instantaneous rate of change of f with respect to x at x_0 is the derivative

$$f'(x_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h},$$

provided the limit exists.

Example 1 The area A of a circle is related to its diameter by the equation

$$A = \frac{\pi}{4} D^2.$$

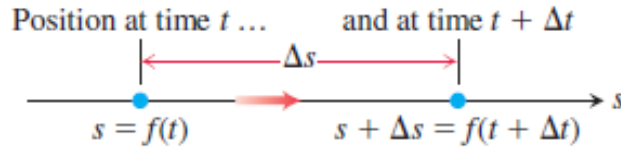
How fast does the area change with respect to the diameter when the diameter is 10 m?

② Motion Along a Line: Displacement, Velocity, Speed, Acceleration

Suppose that an object (or body, considered as a whole mass) is moving along a coordinate line (an s -axis), usually horizontal or vertical, so that we know its position

s on that line as a function of time t :

$$s = f(t).$$



The **displacement** of the object over the time interval from t to $t + \Delta t$ is

$$\Delta s = f(t + \Delta t) - f(t),$$

and the **average velocity** of the object over that time interval is

$$v_{av} = \frac{\text{displacement}}{\text{travel time}} = \frac{\Delta s}{\Delta t} = \frac{f(t + \Delta t) - f(t)}{\Delta t}.$$

To find the body's velocity at the exact instant t , we take the limit of the average velocity over the interval from t to $t + \Delta t$ as Δt shrinks to zero. This limit is the derivative of f with respect to t . Then the velocity (instantaneous velocity) is

$$v(t) = \frac{ds}{dt} = \lim_{\Delta t \rightarrow 0} \frac{f(t + \Delta t) - f(t)}{\Delta t},$$

the speed is the absolute value of velocity

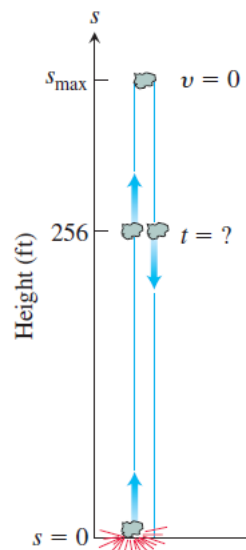
$$\text{Speed} = |v(t)|,$$

and the acceleration is

$$a(t) = \frac{dv}{dt} = \frac{d^2s}{dt^2}.$$

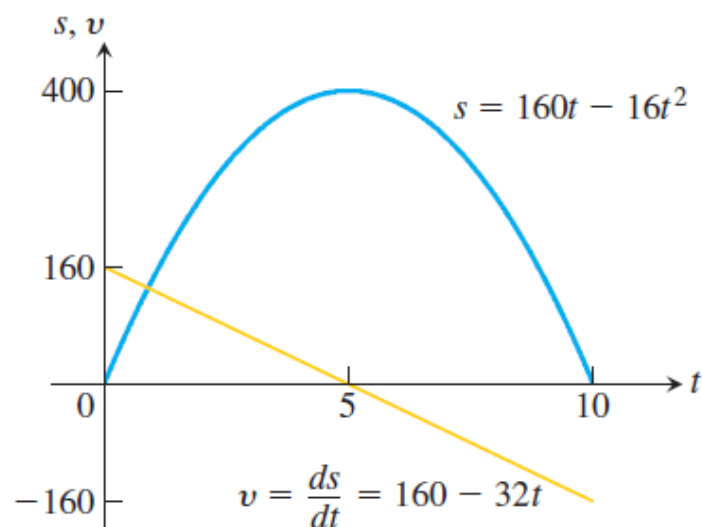
Example 2 The metric free-fall equation is $s = 4.9t^2$. If the free fall of a heavy ball bearing released from rest at time $t = 0$ sec, then

- (a) How many meters does the ball fall in the first 3 sec?
- (b) What is its velocity, speed, and acceleration when $t = 3$?



Example 3 A dynamite blast blows a heavy rock straight up with a launch velocity of 160 ft/sec (about 109 mph). It reaches a height of $s = 160t - 16t^2$ ft after t sec.

- (a) How high does the rock go?
- (b) What are the velocity and speed of the rock when it is 256 ft above the ground on the way up? On the way down?
- (c) What is the acceleration of the rock at any time t during its flight (after the blast)?
- (d) When does the rock hit the ground again.



③ Derivatives in Economics

Economists have a specialized vocabulary for rates of change and derivatives. They call them **marginals**.

In a manufacturing operation, the cost of production $c(x)$ is a function of x , the number of units produced. The marginal cost of production is the rate of change of cost with respect to level of production, so it is dc/dx .

Suppose that $c(x)$ represents the dollars needed to produce x tons of steel in one week. It costs more to produce $x + h$ tons per week, and the cost difference, divided by h , is the average cost of producing each additional ton:

$$\frac{c(x+h) - c(x)}{h} = \text{average cost of each of the additional } h \text{ tons of steel produced.}$$

The limit of this ratio as $h \rightarrow 0$ is the marginal cost of producing more steel per week when the current weekly production is x tons:

$$\frac{dc}{dx} = \lim_{h \rightarrow 0} \frac{c(x+h) - c(x)}{h} = \text{marginal cost of production.}$$

Example 4 Suppose that it costs

$$c(x) = x^3 - 6x^2 + 15x$$

dollars to produce x radiators when 8 to 30 radiators are produced and that

$$r(x) = x^3 - 3x^2 + 12x$$

gives the dollar revenue from selling x radiators. Your shop currently produces 10 radiators a day. About how much extra will it cost to produce one more radiator a day, and what is your estimated increase in revenue and increase in profit for selling 11 radiators a day?